

NAIT Guide to Assessment of Sensory Preferences as part of Adult Neurodevelopmental Assessment

Introduction

In adult services, assessment for Autism, ADHD or other neurodevelopmental conditions should include assessment of sensory preferences. Associated information about fuller neurodevelopmental assessment can be found in the following resource: <https://www.thirdspace.scot/wp-content/uploads/2022/03/NAIT-Neurodevelopmental-Assessment-Workbook-March-2022.pdf>.

Sensory Preferences

Sensory preferences have been described in relation to sound, vision, taste, touch, and smell. They also include vestibular (movement and balance), proprioception (body awareness) or interoception (how our body feels inside). Differences can include hypersensitivity, hyposensitivity, and general sensory overload.

Diagnostic criteria relevant to sensory preferences

Sensory reactivity is one of the core diagnostic criteria for ASD. A mismatch between sensory preferences and the physical and social environments in every day life have been associated with poorer outcomes and distress.

Sensory differences are also important in diagnostic criteria for ADHD, for example in people who are movement seeking, fidgety and for whom the sensory environment affects regulation, attention focus and concentration.

DSM 5 and ICD 11 diagnostic criteria use deficit focussed language and as clinicians we can reframe the language and mindsets we apply when talking with and writing or thinking about neurodivergent people. Some examples are here:

Diagnostic indicators informed through assessment of sensory preferences

Autism Spectrum Disorder	ADHD	Changing to positive language
Stereotypies		
- Motor movements - Use of objects - Speech (echolalia, idiosyncratic)	Fidgets/ squirms	Self-regulation - Uses particular touch or movements of the hands, fingers, body to self-regulate and feel calmer - Repeats sounds, words or phrases for the purpose of emotional regulation - Inhibits the desire to 'stim' in public, order to mask or fit in with societal conventions. May describe using stimming in private only
Insistence on sameness		
- Inflexible adherence to routines - Ritualised patterns of verbal/non-verbal behaviour	- Loses things necessary for tasks/activities - Easily distracted	Predictability and desirability Benefits from predictability of some routines to start, engage in or finish involvement in some aspects of daily life
Restricted/ fixed interests		
- Strong attachment to or preoccupation with unusual objects - Excessively circumscribed or preserved interests	Avoids/dislikes tasks requiring sustained mental effort	Sensory preferences - Sensory preferences are related to interests and motivation - Focusses well when motivated and sensory needs are met (e.g. movement breaks)

Hypo or hyper reactivity to sensory stimuli

- | | | |
|--|--|---|
| <ul style="list-style-type: none"> • Apparent indifference to pain of temperature • Adverse response to specific sounds or texture • Excessive smelling or touching • Visual fascination with lights or movement • Only eating bland food | <ul style="list-style-type: none"> • Difficulty engaging in quiet leisurely activities • Movement seeking or 'on the go' as if driven by a motor | <ul style="list-style-type: none"> • Has the following sensory preferences in different environments ... • Is well regulated when ... |
|--|--|---|

Sensory Assessment

Assessment can include direct observation, environmental assessment, and self or proxy reports. There are benefits and limitations in all available assessment tools, and it is recommended that a variety of assessment approaches are used. Observations in real world situations may support clinicians to make decisions regarding environmental accommodations or adaptations. Interventions should be focussed on enhancing daily functioning and participation in everyday life, at home, at work, in the community and in education.

Occupational Therapists are key members of the team and have expertise in this area, to inform, teach and offer consultation. However, assessment of sensory preferences can be done through self-assessment or by other team members with the right knowledge and skills.

Formal Assessment tools

Common formal assessment tools include:

- Adult Adolescent Sensory Profile [AASP]
[Adolescent/Adult Sensory Profile \(pearsonclinical.co.uk\)](https://www.pearsonclinical.co.uk/adolescent-adult-sensory-profile/)
- Adult Sensory History [ASH]
[Adult/Adolescent Sensory History | The Spiral Foundation](https://www.spiralfoundation.org/adult-adolescent-sensory-history/)

Informal questionnaires

OT practitioners have developed and shared a range of tools online. These can be used by multi-disciplinary neurodevelopmental assessment teams and might support conversations about sensory avoidance and preferences.

- The sensory diet questionnaire contains some American terms but could be adapted for your service. It contains a list of sensory activities which may decrease or prevent stress: [Microsoft Word - Sensory Diet Checklist 2007.doc \(ot-innovations.com\)](https://www.ot-innovations.com/microsoft-word-sensory-diet-checklist-2007.doc)
- The SPD (Sensory Processing Disorder) Checklist attached has the adult questions at the back. [Symptoms Checklist | STAR Institute \(sensoryhealth.org\)](https://www.sensoryhealth.org/symptoms-checklist-star-institute/)
- The Sensory Challenge Questionnaire: [SensoryQuestionnaire.pdf \(sensoryprocessingchallenges.com\)](https://www.sensoryprocessingchallenges.com/sensoryquestionnaire.pdf)
- The Sensory Checklist is available for individual use from: [Sensory Diet Activities - SensorySmarts Autism-Education-Trust-Sensory-Assessment-Checklist.pdf \(birmingham.gov.uk\)](https://www.birmingham.gov.uk/info/20063/autism_education_trust_sensory_assessment_checklist/12722)
- Multidimensional Assessment of Interoceptive Awareness: [Multidimensional Assessment of Interoceptive Awareness | UCSF Osher Center for Integrative Medicine](https://www.ucsf.edu/oshers-center-for-integrative-medicine/multidimensional-assessment-of-interoceptive-awareness)
- [Microsoft Word - Sensory Assessment - LRC \(unl.edu\)](https://www.unl.edu/~lrc/microsoft-word-sensory-assessment-lrc/)

NAIT Sensory Checklist

The Checklist at the end of this guide can be used by all clinicians as part of a neurodevelopmental assessment. It can be completed as a 'self-assessment' or used as part of the clinical interview.

Supporting sensory needs

Understanding sensory preferences can support an individual to anticipate situations where they may feel overwhelmed. These preferences can also be used to develop helpful strategies to assist the person to manage their needs by reducing stimuli they find stressful or to participate in activities that make them feel calmer. Some examples include:

Calming strategies: pressure blanket, sucking a sweet, going for a walk, massage, lavender scent, take a break, subdued lighting, headphones, use stairs instead of lift/escalator, sunglasses, close your eyes for a minute.

Alerting Strategies: jogging/running, jumping jacks/ push ups, eating crunchy food, drinking cold drinks, play music, move around, use perfume, work in a busy office, bright lighting, group activities.

Useful links

Here are some links which might be helpful in understanding sensory preferences and strategies that can help:

Autism Level Up

[Autistic created resources to help us think differently about energy levels and regulation https://autismlevelup.com/category/supports-resources/](https://autismlevelup.com/category/supports-resources/)

NAS 'Too much information' campaign short video clips

- <https://www.youtube.com/watch?v=zWPL3QPD4p0>
- <https://www.autism.org.uk/what-we-do/campaign/public-understanding/too-much-information>
- Processing information <https://www.youtube.com/watch?v=WlxXbONMqNc>
- Sensory Overloads <https://www.youtube.com/watch?v=zWPL3QPD4p0>

Pathway for Eating Disorders and Autism developed from Clinical Experience (PEACE)

The PEACE Pathway website has some useful resources relating to sensory needs and eating: [PEACE Pathway -](#)

Your local Occupational Therapy team may have further useful resources and information.

NAIT Sensory Checklist for adults undergoing neurodevelopmental assessment

Name:

Date of birth:

Date completed:

Everybody has different sensory preferences and sensations they dislike. These can draw us to activities and situations we like and can lead us to dislike or avoid some situations. Understanding sensory preferences can help in identifying strategies that help us to feel calm and well regulated more of the time, in a range of environments, at home, at work, in shops, in education or in other community settings.

How to complete the checklist:

For each type of sensation:

- look at the examples
- Note what you like using this list or your own ideas
- Note what you dislike using this list or your own ideas
- Think about the 'impact' of these sensations on you in your day to day life

You can fill this in yourself or in discussion with someone else

1. Movement

Examples include cycling, running/ jogging, walking/hiking, aerobics, dancing, stretching/ yoga/ Pilates/ Tai Chi, swimming, trampoline, rocking in rocking chair or by yourself, playing an instrument, doodling, re-arranging furniture, gardening, shopping, taking a shower, cleaning, driving, fairground rides, chopping wood, washing the car, skiing/skating, travelling by train/ car/ plane/ boat, being upside-down, learning new activities requiring coordination, sitting still.

Strongly like:

Activities I use to reduce stress/ relax

Strongly dislike

Impact

2. Touch and Temperature

Examples include weighted blanket, massage, eating ice, hot or cold baths/ showers, pottery/clay work, stroking a pet, washing dishes, using a stress ball, fidgeting, twirling hair, going barefoot, getting manicure/ pedicure, washing/ styling hair, cooking/baking, feel of certain fabrics, being

hugged, knitting/crochet/sewing, being in the sun/shade, using powders/ lotions, playing a musical instrument
Strongly like
Activities I use to reduce stress/ relax
Strongly dislike
Impact
3. Auditory/ Listening
Examples include enjoying peace and quiet, sound of water fountain, the sound of a fan, people talking, white noise, music box, wind chimes, singing, humming, whistling, plays/ theatre, live concerts, radio shows, the sound of the sea/ ocean, rain, birds chirping, ticking clock, cat purring, using the telephone, listening to music through headphones, listening to musical instruments, guided relaxation/ meditation, types of music, loud bangs.
Strongly like
Activities I use to reduce stress/ relax
Strongly dislike
Impact
4. Vision/ Looking
Examples include looking at; photos, sunset/ sunrise, snow falling, rain, fish in a tank, autumn leaves, art, bubble lamp, mobiles, waterfalls, clouds, stars, ocean waves, watching sport, films, animals, window shopping, reading, looking through different coloured sunglasses, flowers, picture books, fluorescent lights.
Strongly like
Activities I use to reduce stress/ relax
Strongly dislike

Impact
5. Olfactory/ smelling
Examples include scented candles, essential oils, cologne/ perfume/ aftershave, baking / cooking, coffee, freshly cut grass, flowers, orange/citrus, herbs/spices, chopped wood, pet smells, fresh laundry, scented lotions, incense, herbal tea, mint leaves, rubbish bin, bleach/ cleaning products
Strongly like
Activities I use to reduce stress/ relax
Strongly dislike
Impact
6. Gustatory/ tasting/ chewing
Examples include chewing gum, crunchy foods, sour foods, chewing ice, sucking a thick milkshake through a straw, chewing on your straw, deep breathing, yawning, biting into a lemon, eating a lollipop, drinking coffee/ cocoa/ tea/ herbal tea, eating mints, aniseed balls, eating carrot sticks, spicy foods, ice lollies, blowing bubbles, chocolate.
Strongly like
Activities I use to reduce stress/ relax
Strongly dislike
Impact